Lab 1: ELF - Backward ldd

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GitHub link: https://github.com/versceana/advanced-linux/tree/main/lab1-bldd

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Task

Implement `bldd` (backward ldd) application that recursively scans directories and finds executable ELF files using specified shared libraries.

Implementation steps

Step 1: Project setup

Created modular project structure consisting of four main components:

- `cli.py` - command-line interface
- `scanner.py` - ELF parsing and dependency extraction
- `arch.py` - architecture mapping
- `report.py` - TXT/PDF report generation

Used dependencies:

- `pyelftools`
- `typer`
- `rich`
- `reportlab`

Step 2: Architecture module (`arch.py`)

Implemented `machine_to_arch()` function supporting both integer and string ELF machine identifiers from `pyelftools`.

Supported architectures:

- x86 (`EM_386: 3`)
- x86_64 (`EM_X86_64: 62`)
- armv7 (`EM_ARM: 40`)
- aarch64 (`EM_AARCH64: 183`)

Step 3: Scanner module (`scanner.py`)

Implemented core ELF scanning logic.

`is_executable()`

Detects executable ELF files:

- `ET_EXEC` for standard executables
- `ET_DYN` with `PT_INTERP` for PIE executables
- skips shared libraries without interpreter section

`get_dependencies()`

Extracts `DT_NEEDED` entries from the `.dynamic` section.

`scan_directory()`

Recursively walks through directories, parses ELF files, extracts dependencies, and groups results by architecture.

Non-ELF and unreadable files are skipped safely without interrupting execution.

Step 4: Report module (`report.py`)

Implemented report generation in both TXT and PDF formats.

TXT report example

```
Report on dynamic used libraries by ELF executables on /bin
```

```
---------- aarch64 -----------
libc.so.6 (45 execs)
 -> /bin/bash
 -> /bin/cat
 -> /bin/ls
```

PDF reports

PDF reports are generated using `reportlab` with formatted sections and grouped output.

Generated reports are stored in:

```
reports/generated/
```

Step 5: CLI module (`cli.py`)

Implemented CLI interface using `typer`.

Available options:

- `--scan-dir` - target directory
- `-l, --libs` - shared library names
- `-o, --output` - output report path
- `-f, --format` - report format (`txt` or `pdf`)
- `-v, --verbose` - debug information
- `--help` - usage instructions
- `--version` - application version

Step 6: Testing and Verification

Application was tested inside Linux Docker container on `aarch64`.

Example execution:

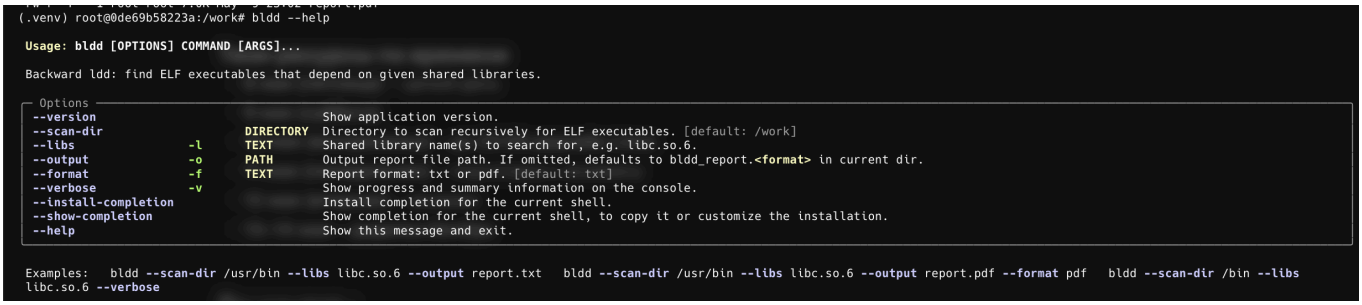
```
$ bldd --scan-dir /bin --libs libc.so.6 --verbose

[DEBUG] printenv: arch=aarch64, deps=['libc.so.6', 'ld-linux-aarch64.so.1']
[DEBUG] rev: arch=aarch64, deps=['libc.so.6']
...
```

```
[DEBUG] Processed 654 files, 483 ELF, 482 matched

Report generated: reports/generated/bldd_report.txt
Total matched library usages: 482
```

Screenshots:



Results:

- 482 executable ELF files using `libc.so.6` were detected
- TXT and PDF reports were successfully generated

Step 7: Multiple libraries and Advanced usage

Tested application with multiple libraries and PDF output generation.

Examples:

```
$ bldd --scan-dir /usr/bin -l libssl.so.3 -l libcrypto.so.3 -f pdf -o
ssl_report.pdf

$ bldd --scan-dir /bin -l libc.so.6 -l libcrypt.so.1 -v
```

Verified that results are sorted by executable usage count in descending order.

Requirements checklist

Requirement	Implementation	Status
Configurable directory	`--scan-dir` option	done
Configurable output	`--output` option	done
Multiple libraries	`-l lib1 -l lib2 ...`	done
Sorted by usage count	`sort_results()`	done
4+ architectures	x86, x86_64, armv7, aarch64	done
TXT reports	`generate_txt_report()`	done
PDF reports	`generate_pdf_report()`	done
Help command	`--help` with examples	done
Modular implementation	separate modules	done

Key technical decisions

1. Modular architecture with separate responsibilities for scanning, reporting, CLI, and architecture mapping
2. Graceful error handling for non-ELF and unreadable files
3. Usage of `pyelftools` for ELF parsing
4. Type-safe CLI implementation using `typer`
5. Architecture detection supporting both integer and string machine identifiers

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Conclusion

The application successfully implements all required laboratory functionality.

The project provides:

- recursive ELF executable scanning
- shared library dependency analysis
- multi-architecture support
- TXT and PDF report generation
- configurable CLI interface

The solution was tested in Linux Docker environment on `aarch64` architecture and produced correct grouped dependency reports.